

Supplementary Notes

Spectral tuning and nanoscale localization of single color centers in silicon via controllable strain

Alessandro Buzzi¹, Camille Papon¹, Matteo Pirro^{2,3}, Odiel Hooybergs¹, Hamza Raniwala¹, Valeria Saggio¹, Carlos Errando-Herranz^{2,3}, and Dirk Englund¹

¹Research Laboratory of Electronics, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139, USA

²QuTech and Kavli Institute of Nanoscience, Delft University of Technology, Delft 2628 CJ, Netherlands

³Department of Quantum and Computer Engineering, Delft University of Technology, Delft 2628 CJ, Netherlands

Contents

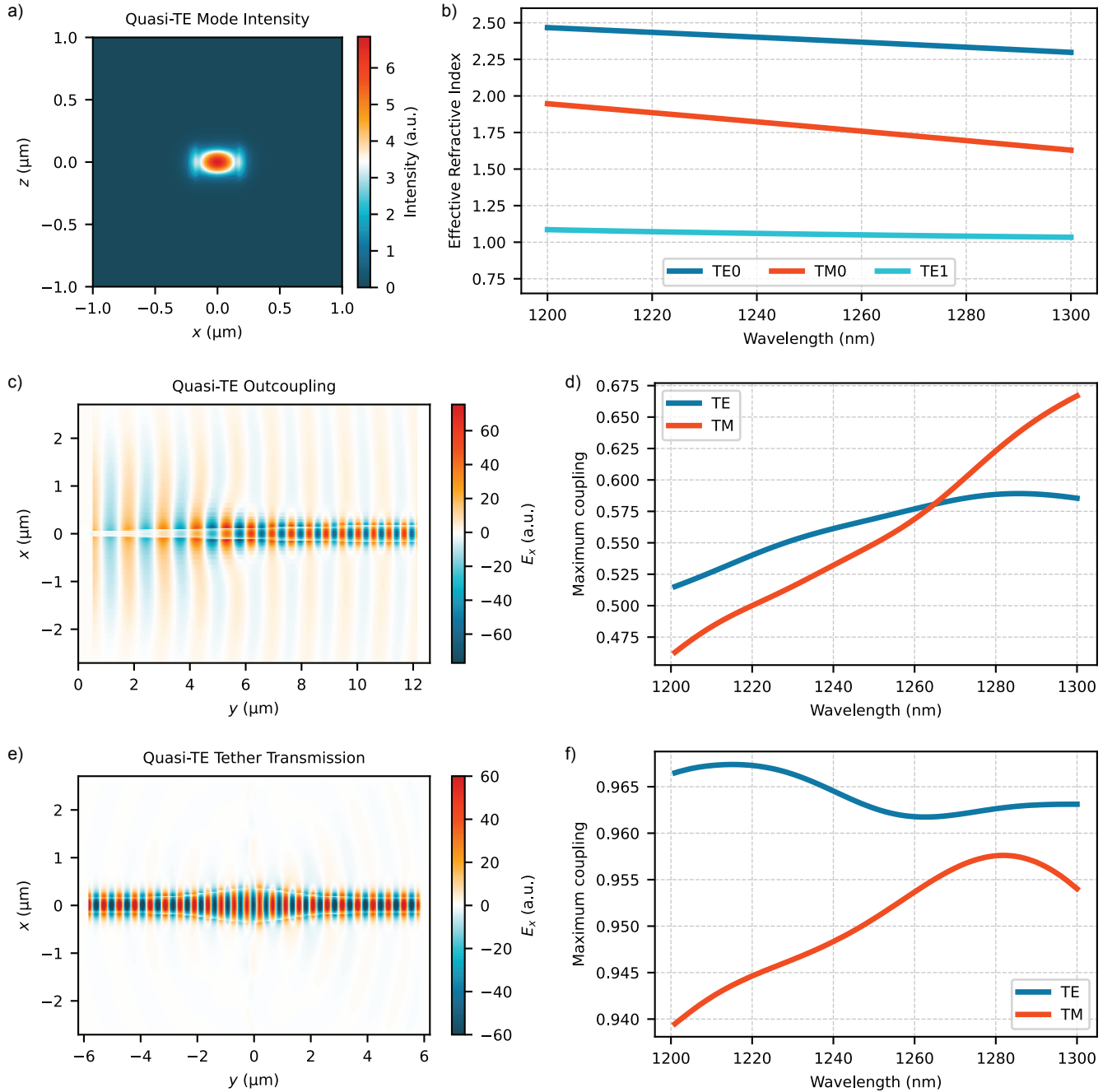
1. Device design	1
2. Experimental setup	3
3. Shadowmask metal evaporation	4
4. Tether loss and coupling efficiency	4
5. Additional spectroscopy results	5
6. UHNA3-SMF28 fiber splicing optimization	8
7. Failure mode of the cantilever actuation	9
8. Secondary Ion Mass Spectroscopy	9
9. Coupling efficiency to the waveguide mode	10
10. Nanoscale localization model for color centers	13
11. Geometry approximation of maximum strain	15
12. Poisson's ratio in the laboratory coordinate system	17
13. Piezospectroscopic model	18
A. Review piezospectroscopic model	18
B. Piezospectroscopic shift with $[\bar{1}10]$ uniaxial strain	19
C. T-center model and comparison	20

Supplementary Note 1: Device design

In this section, we present additional details about the design and simulation of the photonic components of the device. The layout of the photonic structures was created using the Python package PHIDL [1], and the simulations presented here were performed with Tidy3D. The waveguide, which is 350 nm wide and 220 nm thick, supports a fundamental quasi-TE mode, shown in Fig. 1a, and a quasi-TM mode. At a wavelength of 1279 nm, the effective refractive index is 2.34 for the quasi-TE mode and 1.70 for the quasi-TM mode, as shown in Supplementary Fig. 1b.

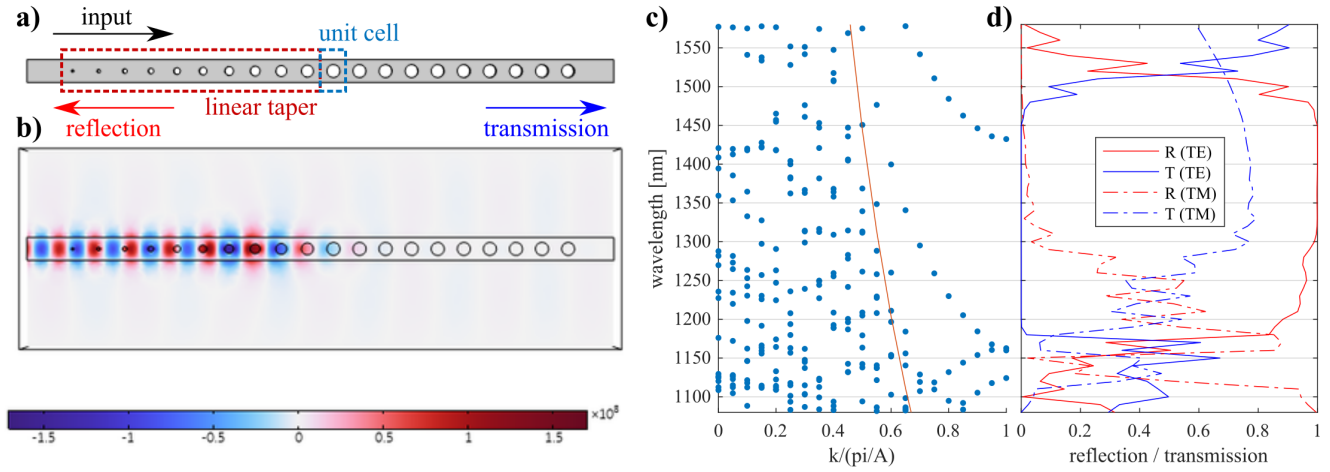
The inverse linear taper, designed to enable efficient mode matching between the Gaussian input source and the waveguide modes, is 10 μm long and tapers to a tip width of 110 nm. Supplementary Fig. 1c illustrates the propagation of the electric field E_x through the taper. The maximum coupling efficiency through the taper at 1279 nm is 59 % for the quasi-TE mode and 62 % for the quasi-TM mode, as shown in Supplementary Fig. 1d.

The tether structure provides mechanical support while minimizing optical losses. Supplementary Fig. 1e shows the electric field E_x as the quasi-TE mode propagates through the tether. At a wavelength of 1279 nm, the transmission through the tether is 96.3 % for the quasi-TE mode and 95.8 % for the quasi-TM mode, as shown in Supplementary Fig. 1f.



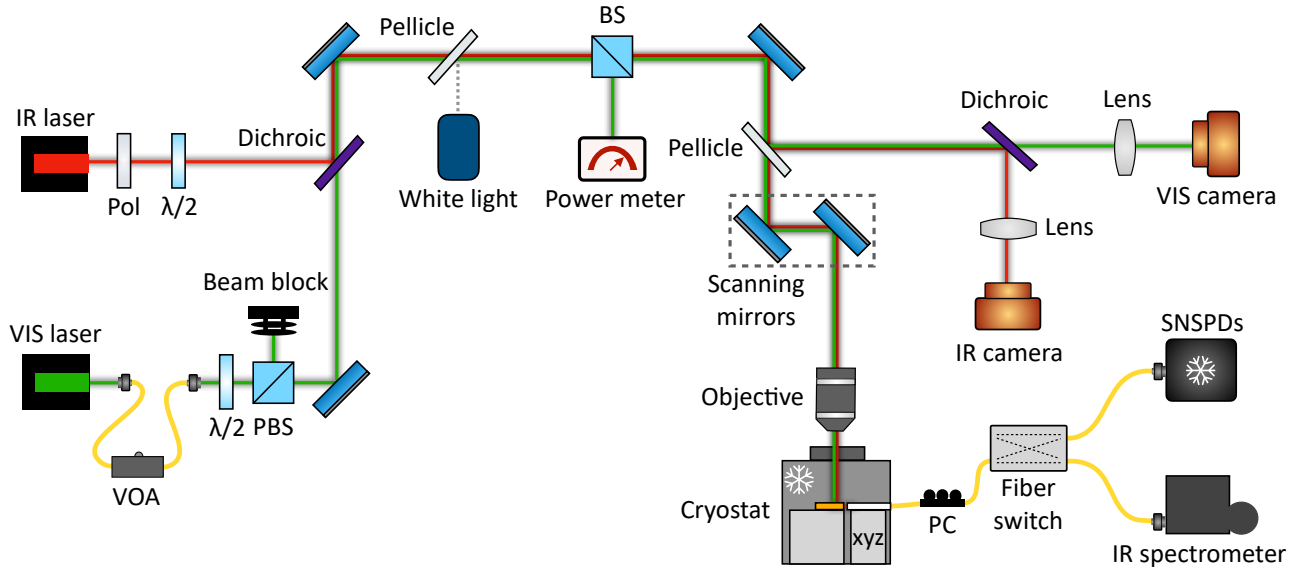
Supplementary Figure 1. FDTD simulations for the design of photonic components. **a)** Mode profile intensity of the quasi-TE mode of the waveguide. **b)** Effective refractive index of waveguide modes over wavelength. **c)** Electric field E_x through the inverse linear taper. **d)** Maximum transmission efficiency from a Gaussian source to waveguide modes. **e)** Electric field E_x through the tether. **f)** Transmission through the tether versus wavelength for waveguide modes.

The reflector provides reflection around 1279 nm with low optical scattering thanks to the linear taper design, shown in Supplementary Fig. 2a. Supplementary Fig. 2c-d shows the reflection and transmission of the reflector around 1279 nm for quasi-TE and quasi-TM modes, agreeing with the band structure of the reflector unit cell. At 1279 nm, the reflection is about 95.35 % for the quasi-TE mode and about 31.5 % for the quasi-TM mode.



Supplementary Figure 2. FEM simulation of the reflector using COMSOL Multiphysics 6.2. **a)** reflector design and **b)** associated y-component of the quasi-TE mode electric field simulation. **c)** bandstructure of the unit cell identified in **a)**. **d)** reflection and transmission of quasi-TE and quasi-TM modes input on the reflector.

Supplementary Note 2: Experimental setup



Supplementary Figure 3. Measurement setup schematic. Our setup consists of a cryogenic confocal microscope equipped with a homemade fiber feedthrough. Infrared (IR) and visible (VIS) laser beams pass through polarization and power control components and are combined at a dichroic mirror to focus onto a sample placed into a cryostat. Their position on the sample is controlled by scanning mirrors, and their focus is set by an objective. A polarizer (Pol) and a half-wave plate ($\lambda/2$) are used on the IR path, while a variable optical attenuator (VOA), $\lambda/2$ and a polarizing beam splitter (PBS) are placed on the VIS path. A beam splitter (BS) is used to read the VIS laser power with a power meter. The sample is placed next to a movable fiber, which collects the PL. The PL is then routed either into superconducting nanowire single-photon detectors (SNSPDs) or an IR spectrometer via a fiber switch, after polarization adjustment via polarization controllers (PC). The IR and VIS reflections from the surface of the sample are imaged on an IR or VIS camera, respectively. White light is used to image the surface of the sample.

Our measurement setup is illustrated in Supplementary Fig. 3. Two continuous-wave (CW) laser beams, infrared (IR) and visible (VIS), are combined at a dichroic mirror to shine onto the sample after passing through a set of scanning mirrors and an objective (Mitutoyo 50 $\times$ M Plan APO). Polarization components such as a polarizer (Pol), a half-wave plate ($\lambda/2$), and a polarizing beam splitter (PBS) are used for polarization and power adjustments. Additionally, a variable optical attenuator (VOA) from Thorlabs is used for automated sweeps of the VIS laser power. A beam splitter (BS) and a power meter are

placed in the setup to read out the VIS laser power. The VIS beam is a Coherent Verdi G5 at 532 nm and is used for G-center excitation, while the IR laser is a tunable laser TSL-570 from Santec set to 1280 nm and serves as a probe for sample alignment. The photoluminescence (PL) originating from the excitation of our waveguide-integrated G-centers is collected into a fiber aligned to the sample waveguide. The ultra-high numerical-aperture fiber is mounted on XYZ cryogenic piezoelectric stages from Attocube and placed next to the sample into the cryostat (Montana Instruments CR-057). After polarization adjustment via polarization controllers (PC), the PL is routed to either superconducting nanowire single-photon detectors (SNSPDs) from Photon Spot or to an IR spectrometer from Princeton Instrument, via the use of a fiber switch from Photonwares. Our SNSPDs feature detection efficiencies of up to 24 %, and are readout with a Swabian Instruments Timetagger 20. Our IR spectrometer consists of a PyLON IR CCD array and two different gratings, one with a density of 300 gr/mm and a 1.2 μm blaze and another with a density of 900 gr/mm and a 1.3 μm blaze. They lead to pixel-defined resolutions of 155 pm and 40 pm, respectively. The IR and VIS reflections from our sample are first separated with a dichroic mirror and then collected with a VIS camera from Thorlabs and an IR camera from Allied Vision, respectively, for laser imaging. White light is used to image the surface of the sample. Excited-state lifetime measurements of our G-centers are performed with a pulsed laser (SuperK from NKT Photonics) with a maximum repetition rate of 78 MHz and filtered by a bandpass filter centered at 532 nm. For PL measurements, a free-space bandpass filtering setup (not shown in the figure), composed of a 1250 nm longpass filter and a 1300 nm shortpass filter, is used to isolate a 50 nm-wide region, including the ZPL. To measure the second-order correlation function and excited-state lifetime, the PL is instead filtered with a tunable wavelength filter from WLPhotonics with a bandwidth of 0.1 nm.

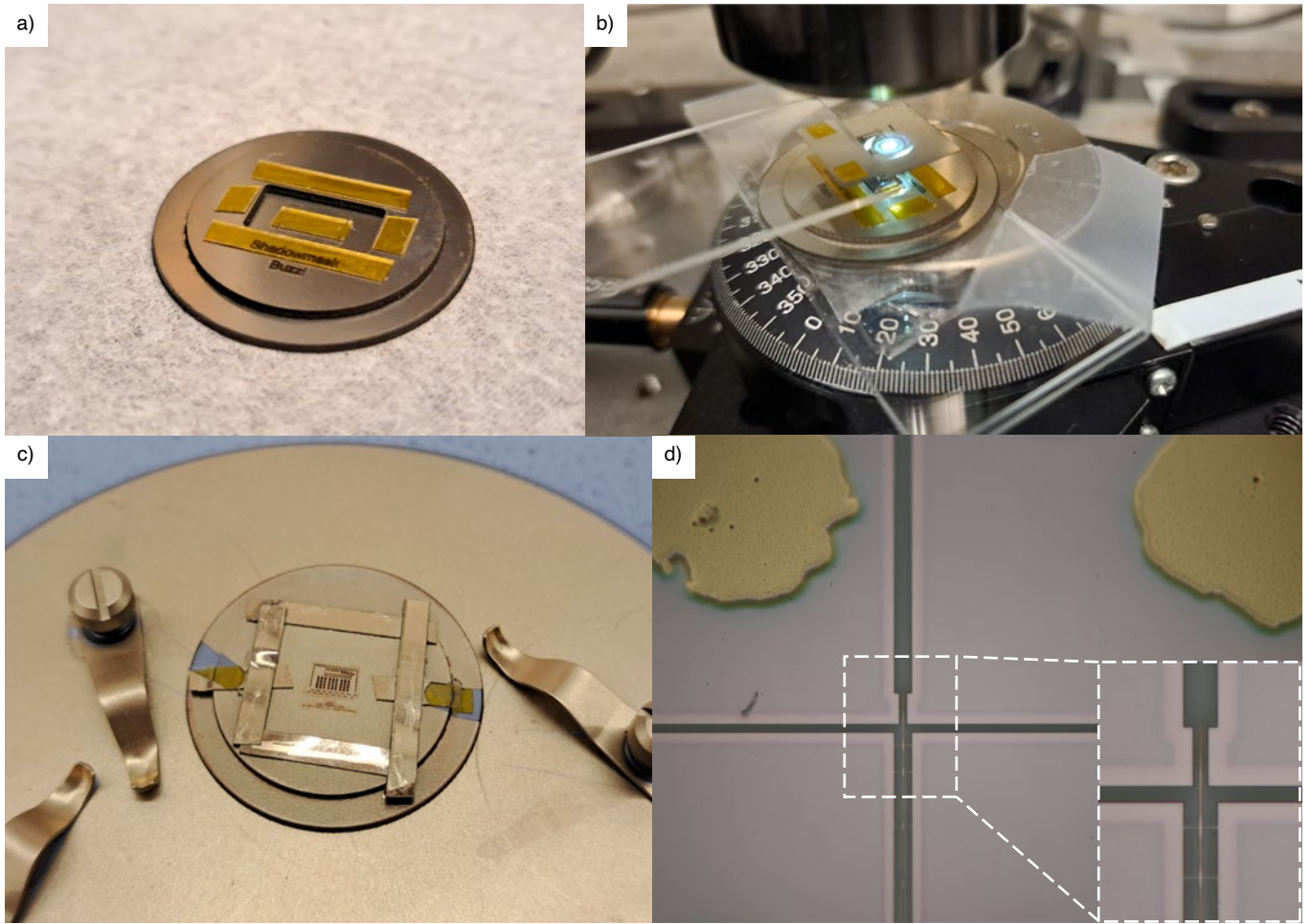
Supplementary Note 3: Shadowmask metal evaporation

In this work, shadow mask evaporation is used to deposit electrical pads directly onto the chip, with chromium (50 nm) and gold (200 nm) serving as the contact materials. This technique avoids both liftoff, which could risk structural failure of the suspended waveguide, and dry metal etching, which may lead to contamination of the sample. By using a shadow mask, we minimize both the risk of structural collapse and potential chemical contamination. The shadow mask was fabricated from a 0.1 mm (4 mil) thick steel sheet using an LPKF ProtoLaser U4 laser cutter at MIT's T.J. Rodgers Laboratory. The laser cutter produces precise patterning, achieving a smallest hole radius of 25 μm , with a minimum spacing of 40 μm between holes. The hole radius used to pattern the electrical pads was 50 μm . The alignment accuracy of the shadow mask on the chip was within 25 μm across the entire chip area (3.7 mm $\times$ 5.2 mm), granting precise positioning of the pads. While very effective for our specific application, it is important to acknowledge that the alignment requirements of this technique would present limitations for achieving tight tolerances or for very large-scale evaporations.

For the evaporation setup, the shadow mask was positioned approximately 100 μm above the chip and held in place by a steel structure (shown in Supplementary Fig. 4a), which stabilized the mask during deposition. After alignment, the holder assembly was placed into an electron beam evaporator (as illustrated in Supplementary Fig. 4c), where the metal pads were deposited via the mask without rotation applied to the chip holder. This configuration prevents pad shapes from broadening, resulting in better edge definition and deposition accuracy. After evaporation, the chip was liberated from its holder. Metal pads were wire-bonded to a printed circuit board before being placed in the experimental setup for characterization.

Supplementary Note 4: Tether loss and coupling efficiency

The tether dimensions of the fabricated devices differ slightly from the designed values and are extracted from an SEM image (Supplementary Fig. 5). Another FDTD simulation using the extracted tether dimensions predicts a transmission efficiency η_{tether} of 94 % for the quasi-TE mode at 1280 nm, shown in Supplementary Fig. 6. Experimentally, the in/out-coupling efficiency is determined by sending 1280 nm light through a loop-back suspended waveguide consisting of 15 tethers. The characterization is done using an array of UHNA3 fibers that are spliced to SMF28 fibers to connect to the experimental setup (cf. Supplementary Note 6). The splice of the input (output) fiber has a transmission efficiency $\eta_{\text{splice,in}}$ ($\eta_{\text{splice,out}}$) of 88 % (80 %). For an input power P_{in} of 1.0 mW the output power P_{out} equals 4.2 μW . Assuming the tether loss is the main contribution to the propagation loss, the coupling efficiency $\eta_{\text{fiber-taper}}$ of the inverse linear taper to the UHNA3 fiber can be extracted from $P_{\text{out}} = P_{\text{in}} \eta_{\text{tether}}^{15} \eta_{\text{fiber-taper}}^2 \eta_{\text{splice,in}} \eta_{\text{splice,out}}$. This results in $\eta_{\text{fiber-taper}} = 12\%$, which combines the taper efficiency, mode matching between the fiber and the taper tip, and Fresnel reflections at the interface. The measured value represents a lower bound since the setup did not allow for optimizing the pitch, roll, and yaw angles.

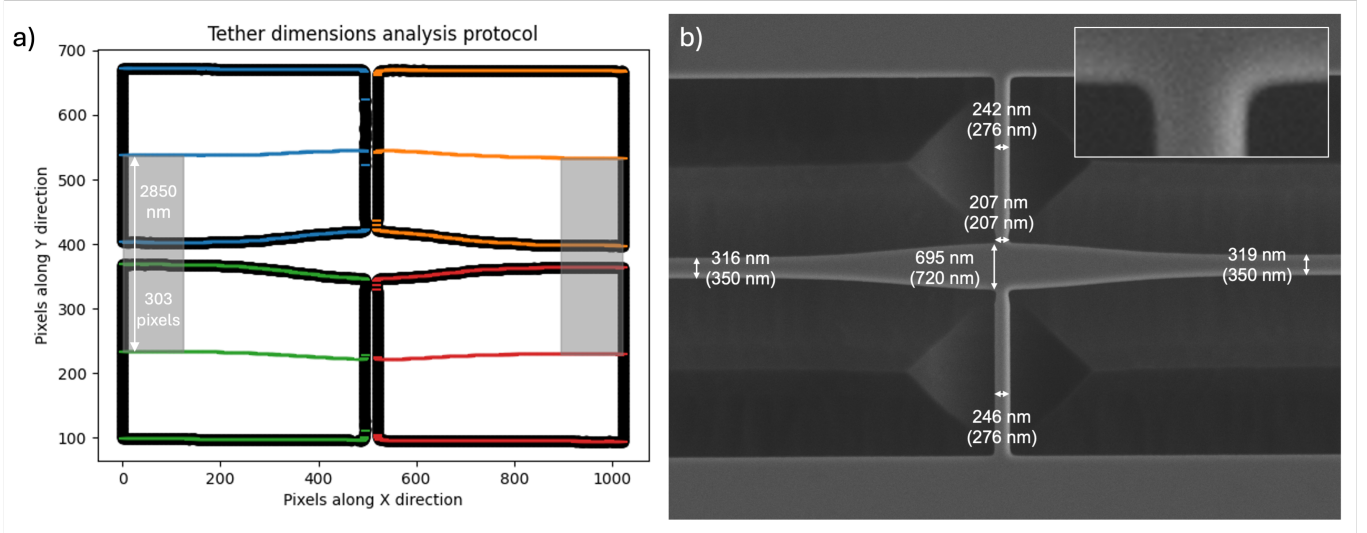


Supplementary Figure 4. Alignment of the steel shadow mask for electrical pad deposition. **a)** Steel structure used to position the mask approximately 100 μm above the chip. **b)** Setup for aligning the steel shadow mask with the chip. The mask is aligned with an optical microscope to ensure that the mask holes are accurately positioned over the designated pad areas on the chip. **c)** Entire assembly with the holder and aligned mask, shown here after the electron beam evaporation of the chromium-gold pads through the mask. **d)** Optical microscope image of a fabricated device, displaying the waveguide cantilever and the metal pads deposited on the chip surface. The inset shows a magnified view of the cantilever structure.

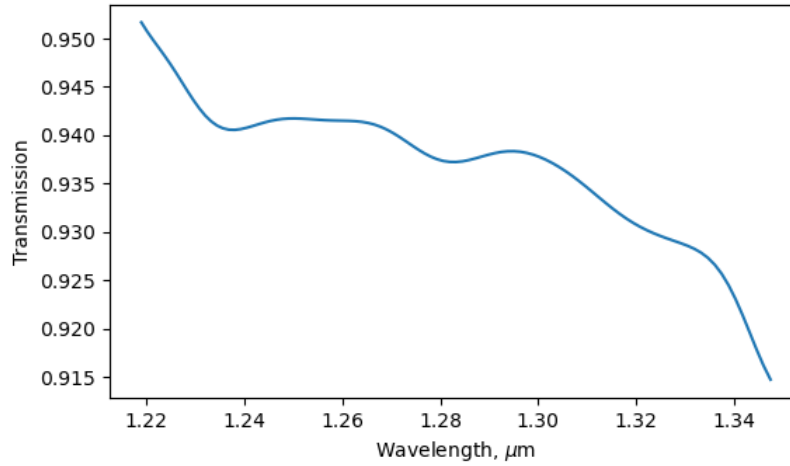
Supplementary Note 5: Additional spectroscopy results

Above-bandgap excitation of G-centers

To demonstrate our ability to excite G-centers in suspended structures, their photoluminescence is characterized in a simple loop-back waveguide, using a 532 nm and 780 nm laser for above-bandgap excitation. The chip containing this device is fabricated in the same way as the sample described in the main text of this paper but with a 10 times lower implantation dose (5×10^{12} ions/cm²) of a different carbon isotope (¹³C) and without electrical pads. The light emitted by the color center is collected using a UHNA3 fiber that is edge-coupled to one of the tapered ends of the suspended waveguide. 2D photoluminescence raster maps are acquired as described in the experimental methods of the main text. The background-corrected emission intensity is extracted for various excitation powers and fitted to a 2-level power saturation model (Supplementary Fig. 7a). The extracted saturation power is 70 ± 19 μW for excitation at 532 nm and 232 ± 11 μW at 780 nm. This discrepancy might arise from different focal spots on the suspended waveguide for both beams and also from the contribution of a higher absorption coefficient at 532 nm compared to 780 nm (approximately 7 times larger). Photoluminescence spectra are characterized below and above saturation power (Supplementary Fig. 7b). The intensity of the emitted light is proportional to the integrated counts of the peak in the spectrum after subtracting the background counts. Taking into account that $I(140 \mu\text{W}) = 0.67 I_{\text{max}}^{532\text{nm}}$ and $I(500 \mu\text{W}) = 0.68 I_{\text{max}}^{780\text{nm}}$ for excitation at the respective wavelengths and powers, we find that $\frac{I_{\text{max}}^{780\text{nm}}}{I_{\text{max}}^{532\text{nm}}} = 1.13 \pm 0.10$, after propagating



Supplementary Figure 5. Tether dimensions. **a)** Protocol to retrieve the actual tether dimensions from an SEM image. The contours are extracted using the cv2 Python module. Due to the working principle of the electron beam lithography and reactive ion etching, the distance between the center of the upper and lower cut-out region can be assumed to be equal to the designed 2850 nm. If too much (or too little) material is removed, this is supposed to happen equally on all edges, and hence, the location of the center is not altered. Averaging over the gray area indicated in the figure, this 2850 nm center separation corresponds to 303 pixels. **b)** SEM image of the tether with indications of the estimated (designed) dimensions. The inset shows the pixel resolution. The pixel size of 9.4 nm is a good indication of the measurement uncertainty in this protocol.

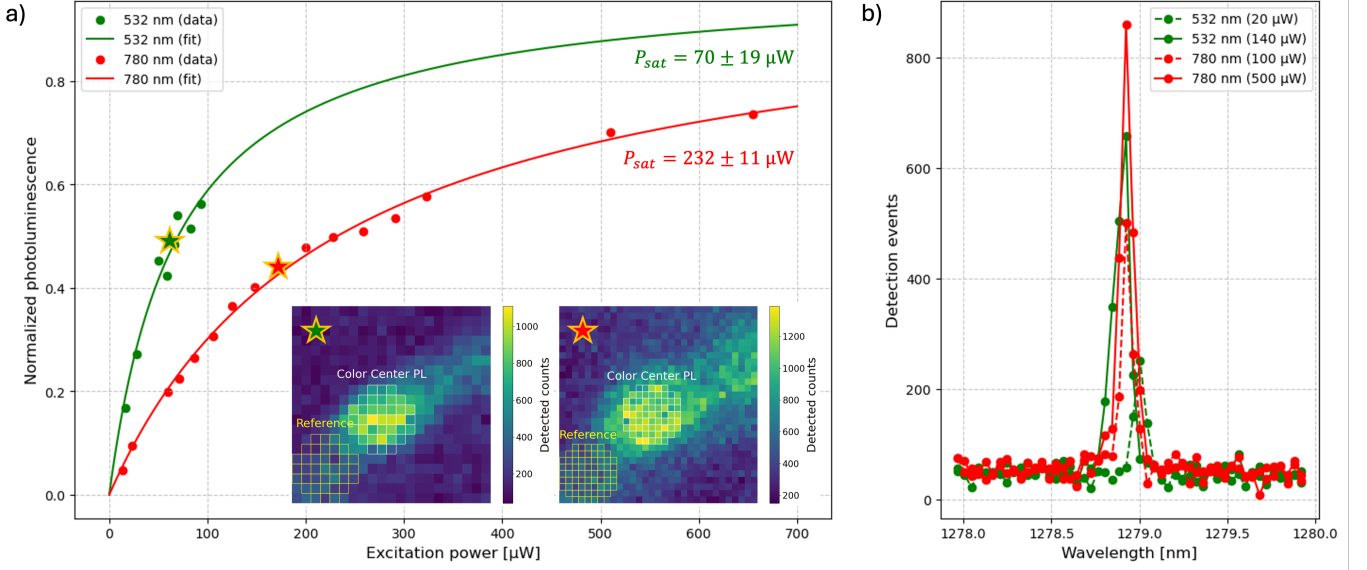


Supplementary Figure 6. FDTD simulation of the transmission of the fundamental quasi-TE waveguide mode. The dimensions are set according to the fabricated and measured values.

errors from the saturation power fit. Since the difference between the calculated ratio and unity is less than 2σ , we conclude that the maximum intensities achieved under the two excitation conditions are not significantly different.

Additional lifetime, second-order correlation and saturation curves

In the following section, we provide additional results from spectroscopy measurements performed on G-centers on localization *A* and *B*, all performed at $\lambda_{exc} = 532$ nm. In Supplementary Fig. 9(a), we provide the time-resolved photoluminescence upon pulsed above-band excitation at low power. The data is fitted to a mono-exponential decay, from which we extract a total decay timescale of $\tau = (6.4 \pm 0.1)$ ns. This value is similar to the value reported in the main text for A_2 and to previous work [2, 3] and has been shown to be excitation power-independent [3]. We further provide the saturation curve of the filtered ZPL A_2

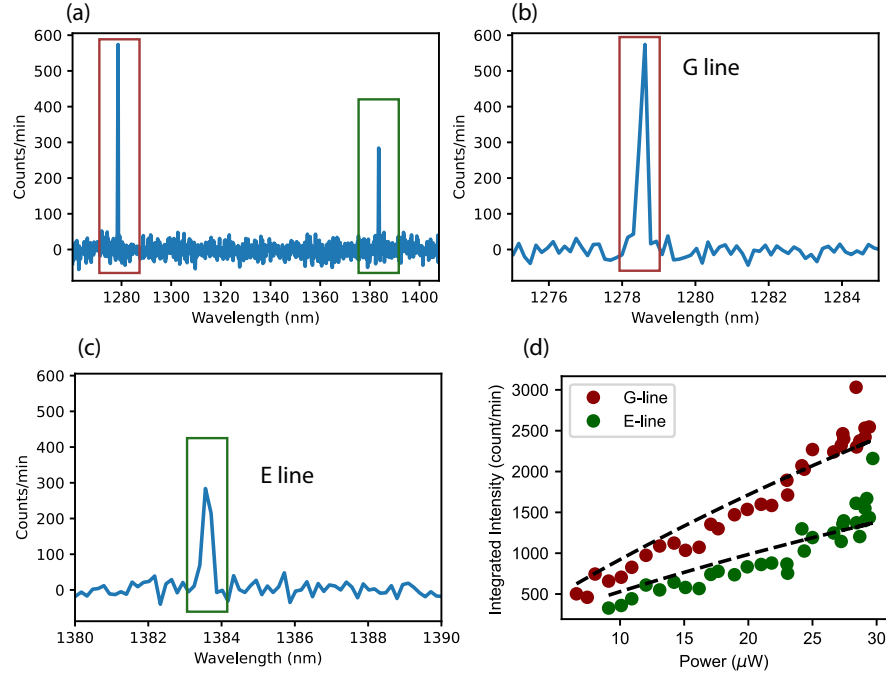


Supplementary Figure 7. Above-bandgap excitation of G-centers with a 532 nm and 780 nm laser. **a)** Power saturation curves. 2D photoluminescence raster maps are made using an SNSPD while sweeping the confocal excitation power (cf. inset of with corresponding stars). The effective photoluminescence count rate is extracted from integrating the counts on the scans around the color center location and subtracting the background counts taken at a reference location in the same waveguide. The background-corrected photoluminescence is fitted to a 2-level emitter saturation model: $I(P) = I_{max} \frac{P}{P + P_{sat}}$ and normalized to the fitted I_{max} . **b)** Spectra of G-center photoluminescence below and above saturation power for both excitation wavelengths.

under pulsed above-band excitation, displayed in Supplementary Fig. 9(b). A fit to the saturation of a two-level system provides a saturation power of $P_{sat} = (3 \pm 2) \mu W$ and a saturated filtered intensity of $I_{inf} = (178 \pm 61)$ counts/s. The lifetime presented in the main text is measured close to saturation. We continue the characterization by measuring the saturation of the same line in continuous-wave excitation, as shown in Supplementary Fig. 9(c), from which we extract a saturation power of $P_{sat} = (11.5 \pm 2.7) \mu W$ and saturated filtered intensity of $I_{inf} = (405 \pm 32)$ counts/s. We then park the laser power around saturation and measure the second-order correlation function, $g^{(2)}(\tau)$, shown in Supplementary Fig. 9(d). Due to the low coincidence count rate, the fit to a three-level system, which includes a bunching term, results in high fitting errors. We then show the fit to the second-order correlation function of a two-level system. The anti-bunching time constant is extracted as $\tau_a = 0.95 \pm 0.65$ ns, however it is not grasping the full physics of the system since slight bunching at short time delays is visible. Nonetheless, we can report an experimentally measured value of $g^{(2)}(0) = 0.12 \pm 0.05$, validating the hypothesis of a single quantum emitter. The error on $g^{(2)}(0)$ is estimated from Poissonian statistics.

Broadband G-center spectra

A specific attribute to genuine G-center is the presence of a red-detuned emission line due to local vibrational mode of the defect [4], around 1381 nm, together with a zero-phonon line around 1278 nm. We record such a broadband spectrum by removing all filtering, shown in Supplementary Fig. 8(a). The spectra are recorded from a waveguide-coupled G-center originating from the same die as strain tuning results obtained in Supplementary Fig. 14. Both emission lines, the zero-phonon line, G-line (Supplementary Fig. 8(b)), and the E-line (Supplementary Fig. 8(c)), originate from the same excited state, we therefore expect similar saturation behavior. We record the spectrum as a function of excitation power and extract the integrated intensity at each peak. We then fit the integrated intensity as a function of power, which yields a similar saturation power for both emission lines, $P_{sat} \approx 122 \mu W$, as shown in Supplementary Fig. 8(d). We note that the energy detuning between the E-line and the ZPL measured here (73.6 meV) differs slightly from reported values in the literature [4, 5] (71.9 meV and 71.6 meV). In our experiment, we report a large intensity ratio between the E-line and ZPL, which resembles experimental values reported in Ref.[4] and is compatible with results from ab-initio calculations [5]. This high intensity ratio may be facilitated by the broadband coupling efficiency, differing from Ref. [6].



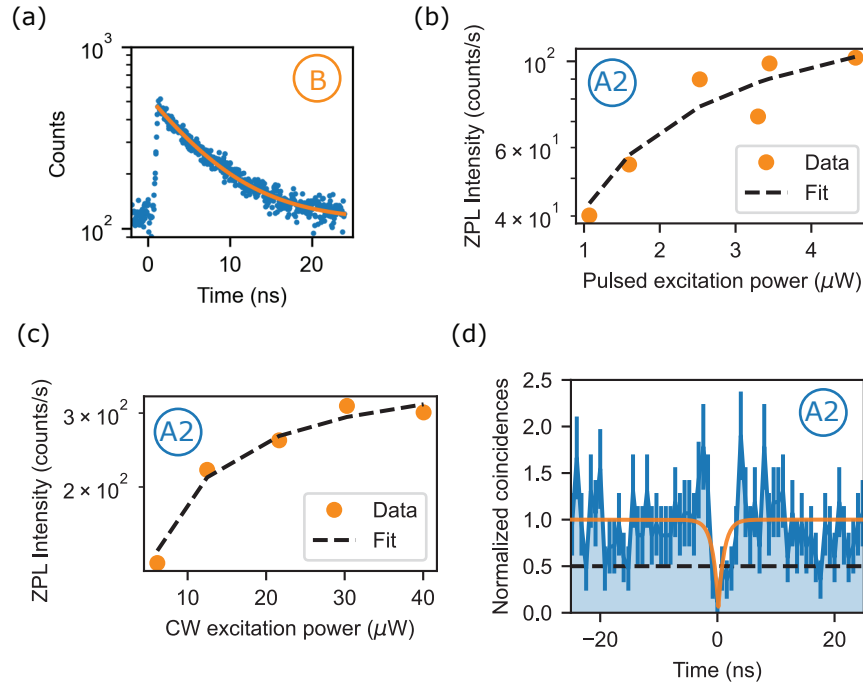
Supplementary Figure 8. (a) Broadband spectrum of waveguide-coupled G-center emission under above-band excitation. The spectrum reveals (b) the zero-phonon line and (c) the E-line, originating from local vibrational modes and expected around 1381 nm. (d) Through power-dependent measurement, we verify that both emission lines originate from the same excited state, by showing similar saturation behavior with $P_{\text{sat}} \approx 122 \mu\text{W}$.

Spectroscopy as a function of tuning voltage

Figure 10 presents the full characteristics of the color centers during the tuning process. Supplementary Fig. 10a shows the intensity, computed as the amplitude of the Lorentzian fit in this case, as a function of voltage. The observed decrease in intensity for lines A_1 , A_2 , and A_3 is attributed to the gradual misalignment of the excitation laser during measurements caused by the relaxation of the galvos. However, the laser spot was optimized at the start of each measurement series to reduce the misalignment issue. Such a drift was not clearly observed for the B line because its spot position was intentionally aligned with the galvos' relaxation position. Supplementary Fig. 10b illustrates the full width at half maximum (FWHM) of the emission peaks versus voltage. Supplementary Fig. 10c depicts the central emission wavelength of the color centers as a function of voltage. Lastly, Supplementary Fig. 10d shows the shift in the central emission wavelength versus voltage. The oscillations visible in Supplementary Fig. 10d, particularly visible for emitter A_2 , result from a small dark region between the pixels of the spectrometer. This phenomenon affects the Lorentzian fitting. From Fig. 3b of the main text, we observe that these oscillations align with the pixel centers of the spectrometer, with dips corresponding to when the emission wavelength is tuned precisely between two pixels. We note that the strain tuning mechanism is reversible, reproducible, and does not affect the linewidth of the emitter. Finally, we provide in Supplementary Fig. 11 the current-voltage map recording during the actuation, which reveals a maximum power dissipated on the order of 10 nW, demonstrating the low-power operation of the device.

Supplementary Note 6: UHNA3-SMF28 fiber splicing optimization

To maximize the coupling efficiency between the taper edge couplers and a single mode fiber, we employ an array of UHNA3 fibers with a mode field diameter of $3.3 \pm 0.3 \mu\text{m}$ at 1310 nm (according to Thorlabs website). Without any additional care, however, the coupling to SMF28 fibers, which constitute the rest of our fiber network, would suffer from loss due to the differing numerical aperture (NA), such that $a = \text{NA}_{\text{UHNA3}}/\text{NA}_{\text{SMF28}} = 0.35/0.14$ leads to a maximal coupling efficiency of $\eta = 4a^2/(1 + a^2)^2 \approx 48\%$. To maximize the coupling efficiency and hence maximize the single-photon count rate, we splice the UHNA3 end fiber from the array with an SMF28 fiber following the recipe provided in Refs. [7, 8]. We optimize the arc time during the splice by measuring the power of a continuous-wave laser at 1280 nm at the output of the fiber array with a power meter head and normalize it by the input power. As shown in Supplementary Fig. 12, the initial splicing efficiency is improved by approximately



Supplementary Figure 9. **a)** Time-resolved measurement of the PL from the color center localized on position *B*. The fit to a mono-exponential decay leads to $\tau = (6.4 \pm 0.1)$ ns. **b)** Filtered ZPL from *A*₂ as a function of pulsed above-band excitation power. The dashed line indicates the fit to power saturation of a two-level system, from which we extract a saturation power of $P_{\text{sat}} = (3 \pm 2)$ μW . **c)** Filtered ZPL from *A*₂ as a function of CW above-band excitation power. From the fit (dashed line), we extract $P_{\text{sat}} = (11.5 \pm 2.7)$ μW . **d)** Second-order correlation function of *A*₂, excited in continuous-wave around saturation, with a measured $g^{(2)}(0) = 0.12 \pm 0.05$. The orange line is a fit for a two-level system. The dashed black line indicates the threshold for interaction with a single emitter.

50 % after further arc splice.

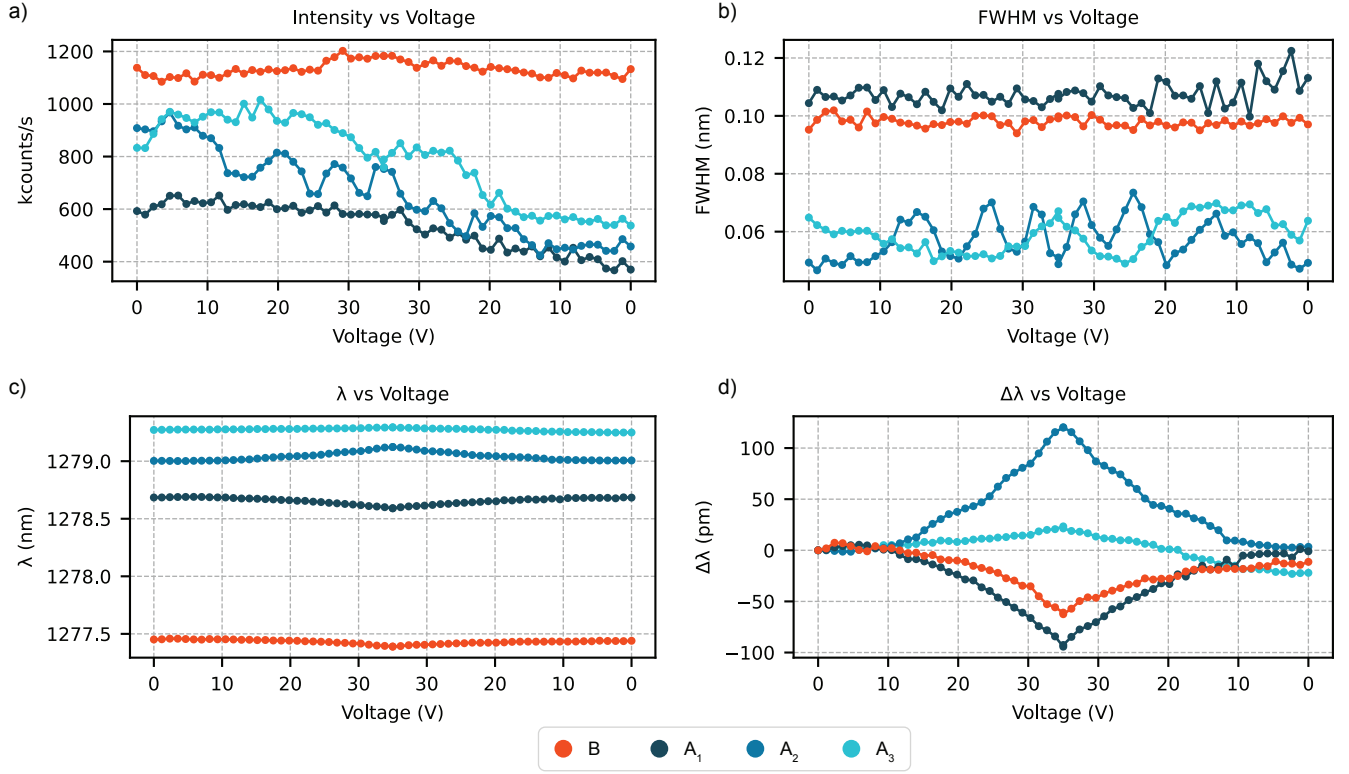
Supplementary Note 7: Failure mode of the cantilever actuation

During the measurement, we observed the failure of the device due to the collapse of a detached part of the lateral pad onto the cantilever, as shown in Supplementary Fig. 13a, where the entire device is visible. The collapse itself is more clearly seen in the inset in Supplementary Fig. 13b, where the portion of the lateral pad that collapsed onto the cantilever waveguide is highlighted, causing structural damage and leading to the device's failure. Supplementary Fig. 13c shows a simulation of the cantilever's maximum displacement as a function of applied voltage, illustrating the transition from 0 V to the pull-in voltage (42.5 V).

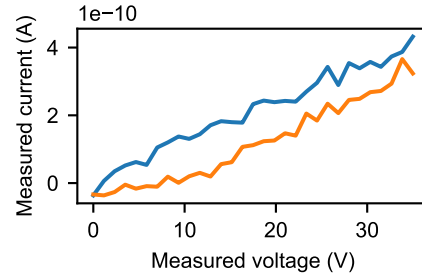
In a different device, with a cantilever length of 15 μm , shown in Supplementary Fig. 14, the heatmap (Supplementary Fig. 14a) illustrates the tuning of the emitters as a function of the vertical driving voltage. At 12.5 V, the right cantilever collapsed onto the central part of the lateral pad (Supplementary Fig. 14b). Initially, the emitters tuned as anticipated, but after the collapse, the device exhibited approximately linear tuning at higher voltages due to the continued possible driving of the collapsed cantilever. The failure occurred because the pad was left floating, and likely, the charging exceeded the lateral pull-in voltage, causing the collapse. Interestingly, a large tuning exceeding 300 pm was observed at 12.5 V. Moreover, we can see from the heatmap in Supplementary Fig. 14a that two color centers on the right were tuned at the same frequency before the collapse, at approximately 10 V. The presence of two distinct emission lines after the failure confirms that the tuning observed prior to collapse originated from two separate emitters.

Supplementary Note 8: Secondary Ion Mass Spectroscopy

Secondary ion mass spectroscopy (SIMS) was performed by Eurofins EAG Materials Science, LLC after annealing and chip fabrication on an edge piece from the same sample as the devices. The results for carbon, silicon, and oxygen concentrations as a function of depth are shown in Supplementary Fig. 15. The increase in oxygen concentration reveals the interface between



Supplementary Figure 10. Characteristics of the color centers during tuning (voltage up and down). **a)** Intensity extracted from a fit to a Lorentzian as a function of applied voltage. **b)** Full width at half maximum (FWHM) of the Lorentzian fit as a function of voltage. **c)** Color center central emission wavelength versus voltage. **d)** Shift in color center central emission wavelength versus voltage.

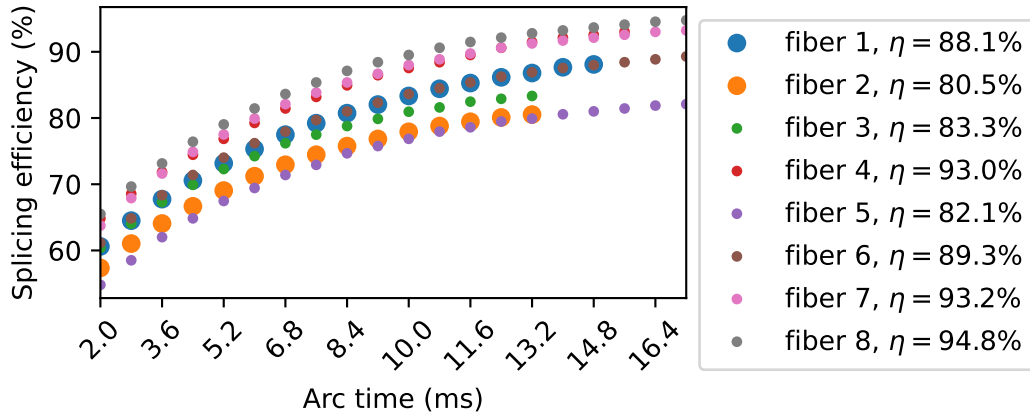


Supplementary Figure 11. Measured current as a function of the applied voltage between the cantilever and the substrate, recorded during the tuning of the color centers. The blue curve is the forward sweep, while the orange curve is the backward sweep. The maximum power dissipated in this operation is ≈ 10 nW.

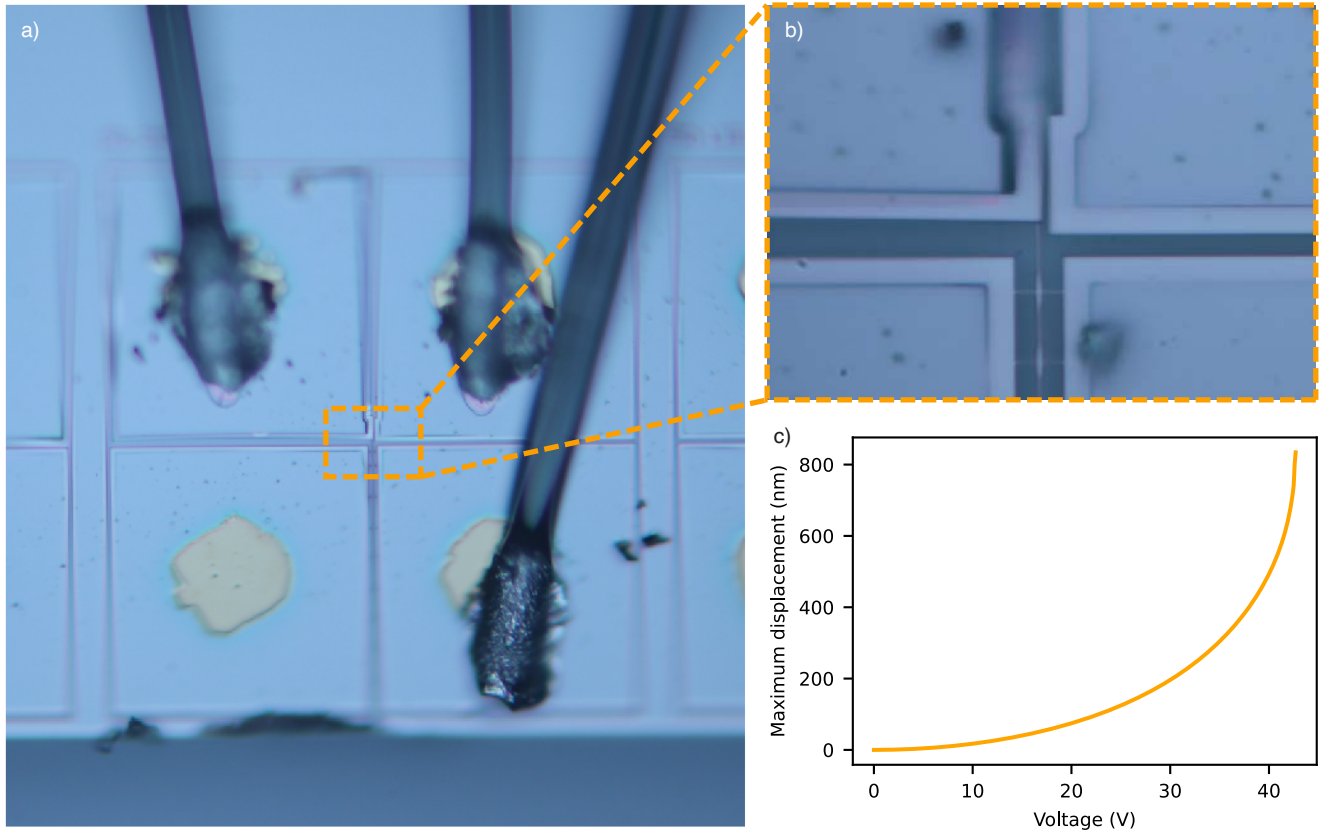
the silicon device layer and the bottom silicon dioxide. The nanoscale localization model uses the SIMS data, considering only the carbon concentration in the central region of the waveguide thickness. Carbon concentrations at the surface and interfaces are excluded, as they are likely due to contamination during processing and do not contribute to the formation of G-centers. As described in the Methods section, the distribution of G-centers is assumed to be proportional to the square of the carbon concentration in the central region of the waveguide.

Supplementary Note 9: Coupling efficiency to the waveguide mode

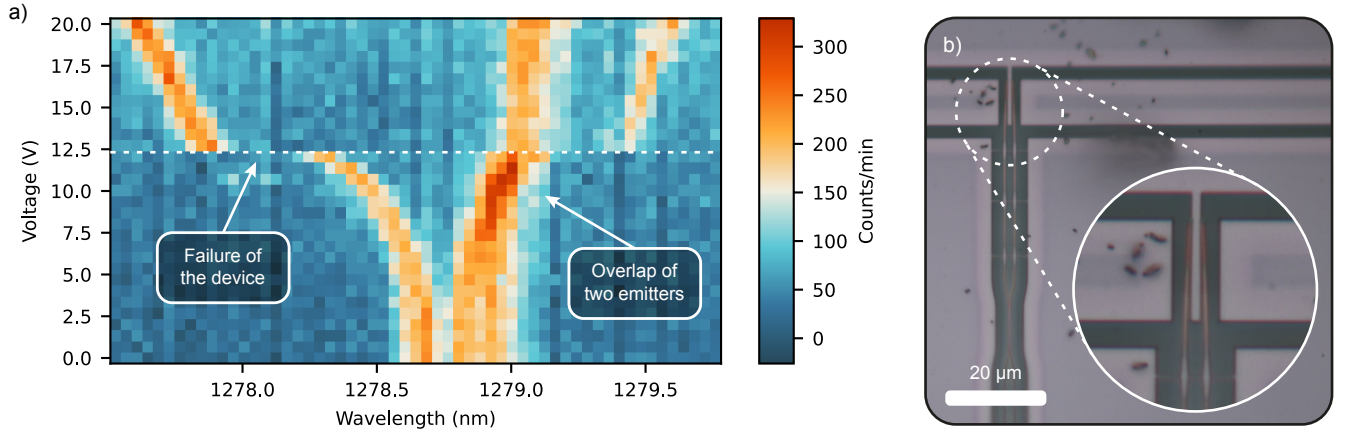
The G-centers can be found with three different classes of dipole orientations. The dipole coupling to the waveguide mode described in Fig.4(f) in the main text is quantified by the β -factor, which is the ratio between the power coupled to a given



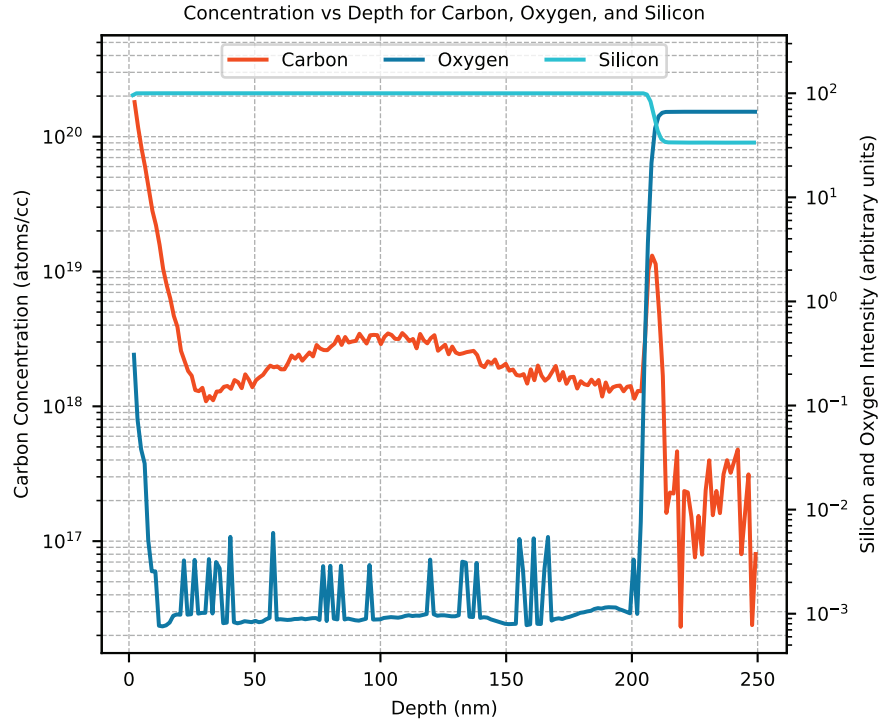
Supplementary Figure 12. Splicing optimization between the UHNA3 array fibers and SMF28. The transmission efficiency is monitored for each extra arc. The fibers used in this work are highlighted with larger markers (fiber 1, fiber 2). The final transmission efficiencies after splicing are indicated in the legend.



Supplementary Figure 13. Post-failure device structure. **a)** Image of the entire device, showing the deposited electrical pads and wire bonds on the chip. The device is observed after the mechanical failure. **b)** Optical microscope close-up of the cantilever waveguide. The image highlights the cause of the failure, revealing a portion of a cracked lateral pad that has collapsed onto the cantilever waveguide, leading to structural damage. **c)** FEM-simulated displacement of the cantilever waveguide as a function of applied voltage, showing the curve from 0 V to the pull-in voltage (42.5 V).



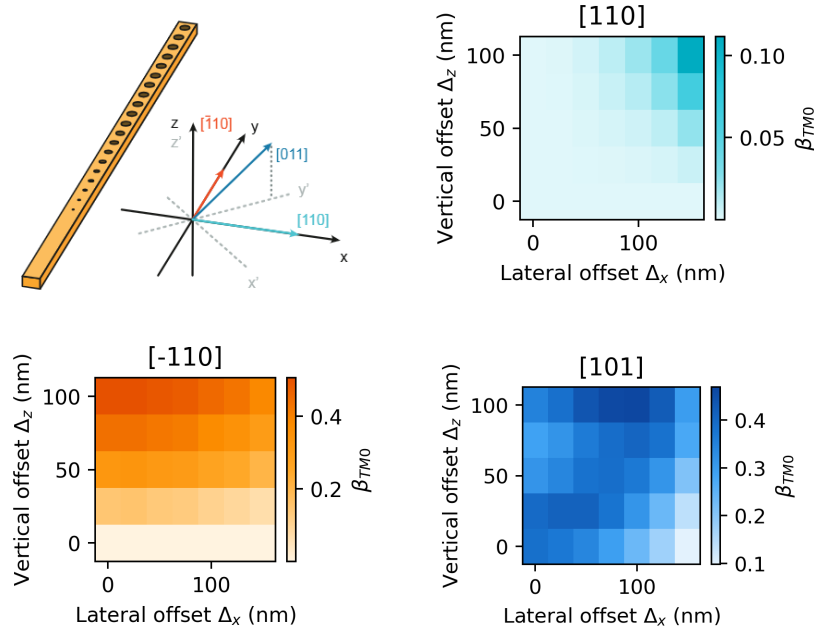
Supplementary Figure 14. Tuning of emitters up to device failure. **(a)** Heatmap showing the emission spectrum as a function of driving voltage. At 12.5 V, the device collapses laterally. **(b)** Optical microscope image of the device after failure, illustrating the cantilever lateral collapse.



Supplementary Figure 15. Secondary ion mass spectroscopy (SIMS) data of the sample, displaying the carbon concentration alongside silicon and oxide concentrations as a function of depth.

waveguide mode and the total power radiated by the dipole. Following the method in Ref. [9], we calculate the power radiated in the fundamental quasi-TE waveguide mode by performing a 3D FDTD [10] calculation carried out using MEEP open-source software package [11]. To do so, we define a single-mode waveguide, which is 350 nm wide and 220 nm tall. The position of a radiating dipole, either oriented along the $[110]$, $[\bar{1}\bar{1}0]$ or $[011]$, is swept across the waveguide width and height, as shown in Fig. 4c-f in the main text. The power coupled into the quasi-TE₀ mode, whose mode profile is shown in Supplementary Fig. 1(a) from Supplementary Note 1, is simulated and gives the β -factor for each equivalence class of dipoles. We assume unidirectional emission thanks to the presence of the reflector, optimized for reflection of the quasi-TE₀ mode (Supplementary Note 1, Supplementary Fig. 2). In the main text, we refer to this simulated quantity as the dipole coupling efficiency to illustrate the probability of detecting each dipole class given its position. This is a more accurate description of the value, since the

β -factor is usually used to quantify the portion of quantum emitter radiation participating in coherent effects. However, in the case of G-centers and other color centers in silicon, the Debye-Waller, quantifying the amount of radiation emitted in the zero-phonon line instead of the phonon side band, is only of $F_{DW} = 0.15$ [12] and will therefore represent an upper bound to the β -factor in the waveguide. Additionally, we compute the coupling into the fundamental quasi-TM mode of the waveguide for each dipole class, as shown in Supplementary Fig. 16. We note that the $[110]$ dipole, due to its transverse orientation to the TM electromagnetic field distribution, couples poorly to this waveguide mode. Moreover, the $[\bar{1}10]$ dipole also shows low coupling to the TM waveguide mode, especially for small vertical offset. The value reported in Supplementary Fig. 16 consider unidirectional emission, however the total coupling efficiency is weighted by the low reflection efficiency of the reflectors for the TM mode, as shown in Supplementary Fig. 2. To get the effective coupling efficiency, we need to apply a factor of 65 % to simulated values, leading all efficiencies to be less than 20 %, further less if we restrict to dipoles located in central areas of the waveguide. Since the coupling efficiency to the TE waveguide mode is larger than to TM mode for all dipole classes, in the waveguide region of higher G-centers probability, we only consider the coupling to TE mode in the full model presented in Fig.4 in the main text.



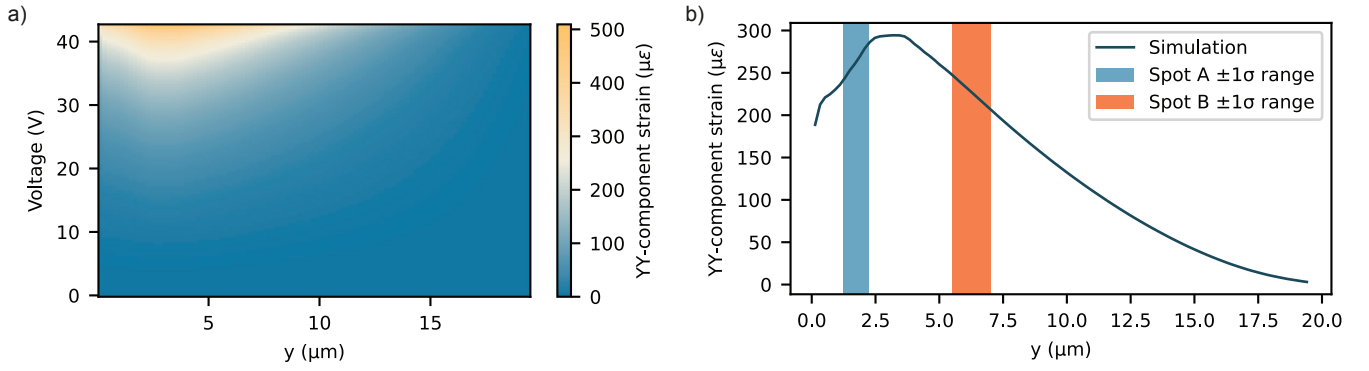
Supplementary Figure 16. FDTD simulation of coupling efficiency to the quasi-TM waveguide mode for the three different classes of dipoles, as indicated in the schematics.

Supplementary Note 10: Nanoscale localization model for color centers

The nanoscale localization model estimates the vertical positions and orientations of color centers within the cantilever waveguide. This supplementary note provides further details on the FEM simulations, the maximum likelihood estimation (MLE) of generation rates, and the Monte Carlo simulations. Additionally, we present the probability distributions for different scenarios, comparing results obtained with and without incorporating MLE for the color centers' orientations.

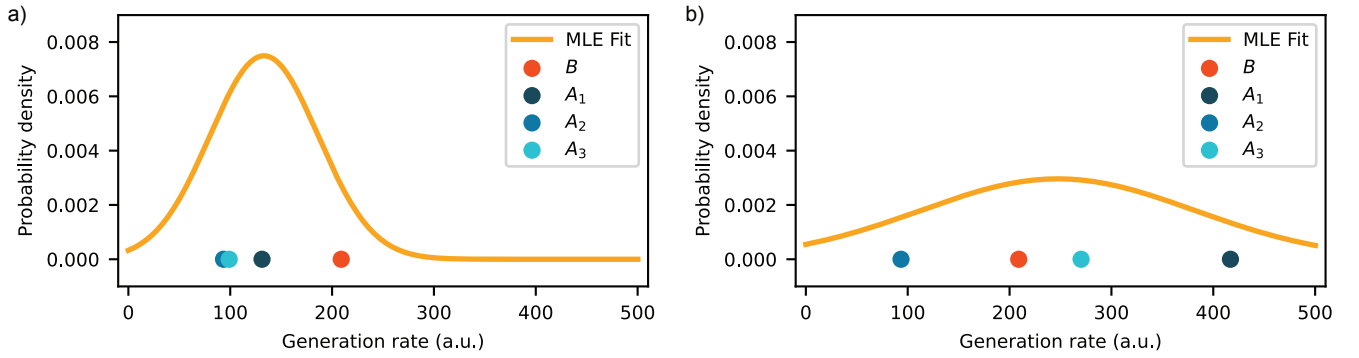
FEM simulation and strain distribution

The strain distribution along the cantilever waveguide was simulated using the electromechanical finite element method in COMSOL Multiphysics. The results are shown in Supplementary Fig. 17a, where the heatmap illustrates how the maximum strain varies with different driving voltages along the cantilever. Supplementary Fig. 17b presents the strain profile at a driving voltage of 35 V, highlighting the spatial positions of the color centers.



Supplementary Figure 17. Strain distribution simulated by FEM along the cantilever. **a)** Heatmap of the strain distribution along the cantilever at different driving voltages. **b)** Strain profile along the cantilever at 35 V, highlighting the spot positions corresponding to the color centers' positions along the cantilever.

Maximum likelihood estimation of generation rates



Supplementary Figure 18. Maximum likelihood estimation of the generation rate for two scenarios with different dipole orientations of color centers. **a)** Estimation for the scenario where all emitters are aligned to the 0 equivalence class of orientations, corresponding to a more likely configuration. **b)** Estimation for the scenario where emitters A_1 and A_3 are aligned to the 1 equivalence class of orientations, representing a less likely configuration.

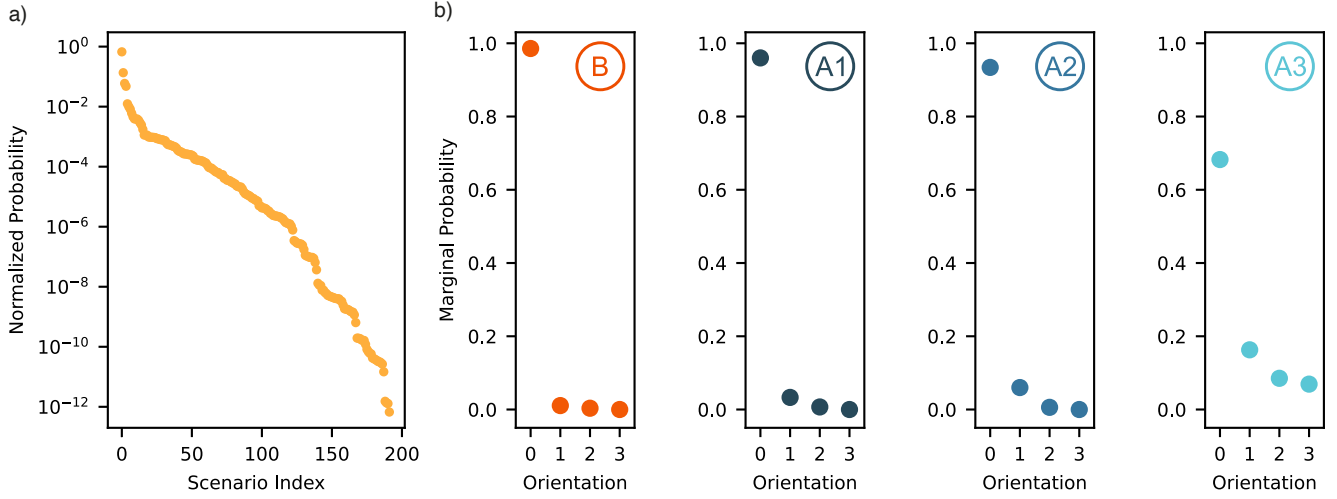
As described in the main text, the generation rates for each emitter are estimated using MLE applied to Gaussian distributions. The MLE is performed on the estimated intensities of the emitters after accounting for coupling. The resulting values are proportional to the generation rates of the emitters. The intensities are computed by integrating the area under the Lorentzian fit of each peak and averaging over ten successive measurements. Supplementary Fig. 18 illustrates examples of likely and unlikely generation rate distributions based on this analysis.

Probability distributions for vertical positions and orientations

To evaluate the probabilities of emitters occupying specific vertical positions and orientations, we calculate the product of two probabilities:

1. The probability of an emitter being at a certain vertical position given its orientation, based on SIMS carbon concentration data.
2. The MLE-derived probabilities of color centers aligning along specific dipole orientations.

Supplementary Fig. 19 displays both the joint and marginal probability distributions for the emitters' orientations and corresponding vertical positions. The joint distributions (Supplementary Fig. 19a) represent multiple orientation scenarios occurring simultaneously, while the marginal distributions (Supplementary Fig. 19b) depict probabilities for individual orientations. The



Supplementary Figure 19. Probabilities of different dipolar orientation scenarios for each emitter. **a)** Joint probability distribution for the scenarios. **b)** Marginal probabilities for each emitter.

orientation classes considered in the model are listed as classes 0 to 3, corresponding to the classifications in Supplementary Note 13: F_1 , F_2 , F_3 – F_6 , and F_4 – F_5 .

Monte Carlo simulations

Monte Carlo simulations were performed to quantify the uncertainty in vertical localization under the most likely orientation scenarios. The input distributions for the emitters' vertical positions are shown in the left panels of Supplementary Fig. 20, while the output distributions, incorporating Gaussian errors from FEM simulations (20 %) and piezospectroscopic constants (4.7 %), are displayed in the right panels. The mean and standard deviations of these distributions are reported at the top of each plot. Panels a) to d) refer to individual emitters B, A₁, A₂, and A₃, respectively.

Comparison of scenarios

Supplementary Fig. 21 compares the probabilities of different scenarios. The left side of Supplementary Fig. 21a presents probabilities derived from SIMS carbon concentration data only, assuming equal likelihood for all emitter orientations. The right side of Supplementary Fig. 21b integrates both the carbon concentration and the MLE for the emitters' orientations, as evident from these maps, incorporating the MLE results in sharper and more precise estimates of the vertical positions of the color centers.

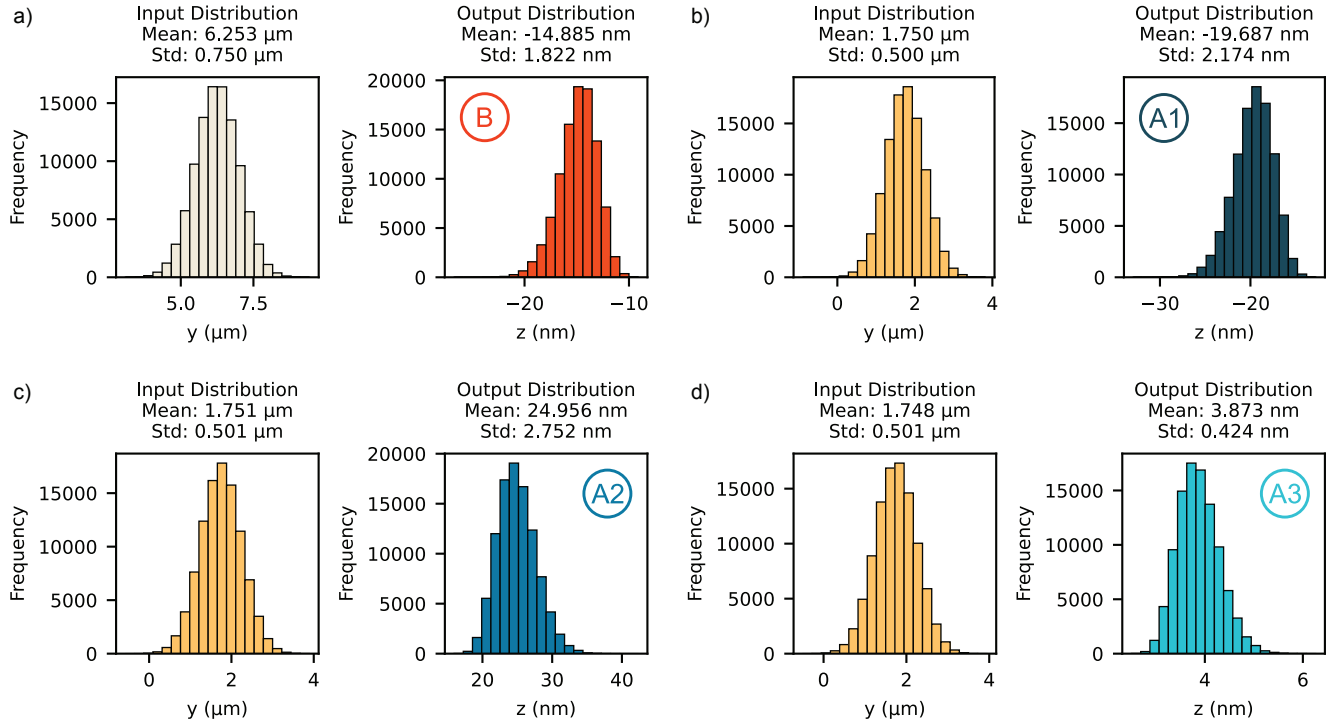
Supplementary Note 11: Geometry approximation of maximum strain

To estimate the maximum strain achievable with the cantilever structure proposed in this work, the Euler–Bernoulli beam theory can be used under the assumption that the small-strain condition is valid, as commonly done in MEMS cantilever models [13, 14]. Alternatively, the same result can be obtained through a simple geometric argument. Referring to the structure depicted in Supplementary Fig. 22, R denotes the radius of curvature, θ the angle subtended by the curvature of the cantilever, L the unstrained length of the cantilever, L' the strained length (at the top surface of the cantilever), d the vertical displacement, and t the thickness of the cantilever. The maximum strain on the upper side of the cantilever is given by

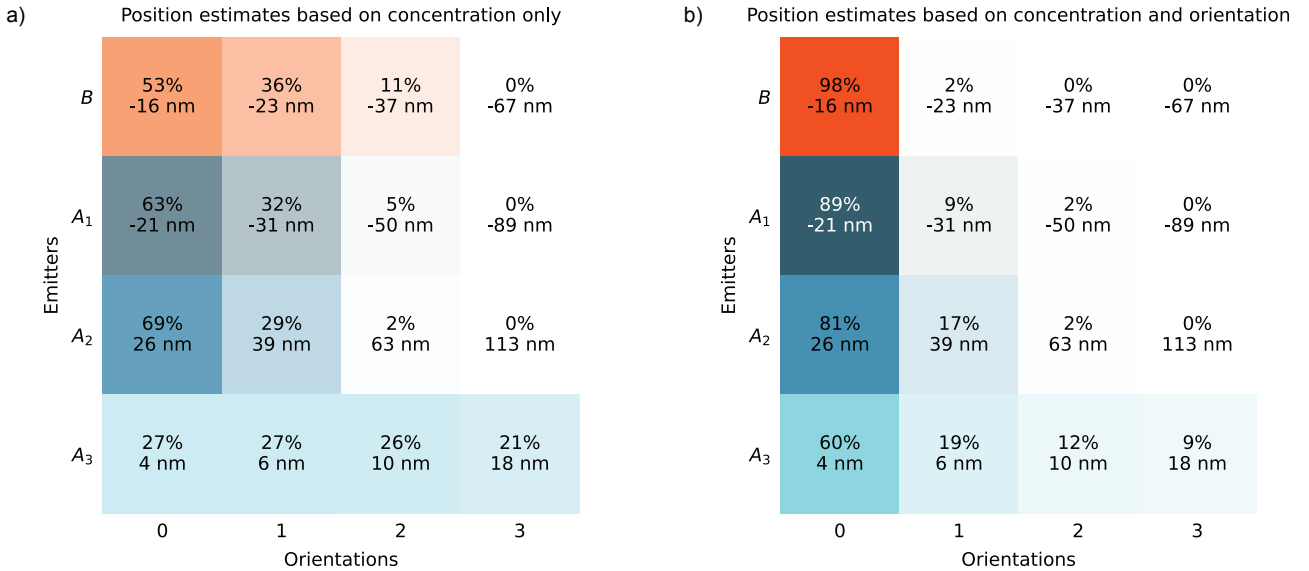
$$\epsilon_{\max} = \frac{\Delta L}{L} = \frac{L' - L}{L} = \frac{L'}{L} - 1.$$

By substituting $L' = (R + t/2)\theta$ and $L = R\theta$, we find

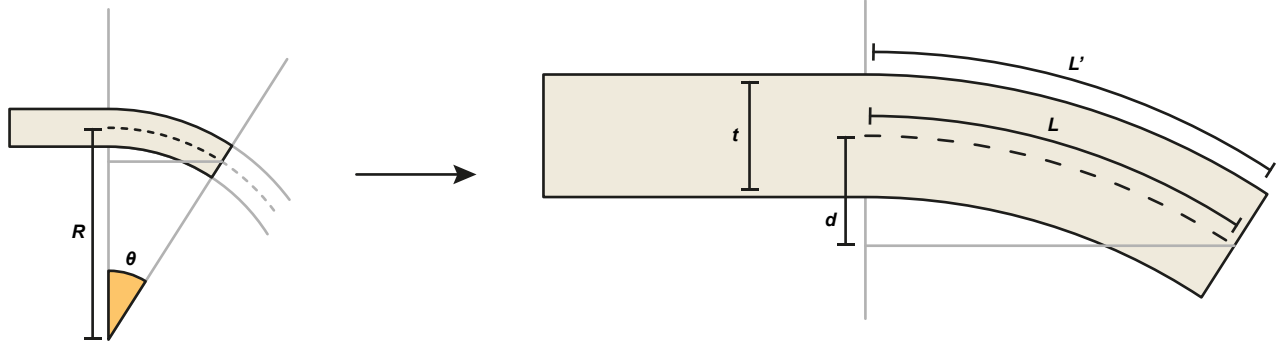
$$\epsilon_{\max} = \frac{t}{2R},$$



Supplementary Figure 20. Results of Monte Carlo simulations estimating the error in the vertical localization of emitters. For each emitter, the left panels display the input probability distribution for the emitter's position along the cantilever, determined by the excitation spot position and size. The right panels show the output probability distribution, incorporating Gaussian errors of 20 % error from FEM simulations and 4.7 % error from piezospectroscopic constants. The mean and standard deviations of the distributions are reported on top of the plots. **a)** Emitter B. **b)** Emitter A₁. **c)** Emitter A₂. **d)** Emitter A₃.



Supplementary Figure 21. Distribution of probabilities for each vertical position of the color centers. **a)** On the left are the probabilities associated with each position, considering only the information based on the carbon concentration. **b)** On the right, the map shows the probabilities associated with considering both the carbon concentration and the probabilities given by the maximum likelihood estimation of the color centers' orientations.



Supplementary Figure 22. Lateral (XZ) cross-section schematic illustrating the geometrical parameters of the bent cantilever waveguide.

which corresponds to the result obtained with the Euler-Bernoulli theory. To express R as a function of displacement d , we use $d = R(1 - \cos \theta)$. For small θ , $d \sim R\theta^2/2$, yielding $\theta = \sqrt{2d/R}$. With $\theta = L/R$, we derive $R = L^2/(2d)$. Thus, $\epsilon_{\max}$ becomes

$$\epsilon_{\max} = \frac{td}{L^2}.$$

Using d as the gap between the device layer and the silicon handle, we calculate a maximum strain of approximately $1.1 \text{ m}\epsilon$. If d is taken as the pull-in displacement, equal to a third of the gap according to the parallel plate capacitor model [13], $\epsilon_{\max}$ reduces to $367 \text{ }\mu\epsilon$, which closely aligns with finite-element-method simulations.

Supplementary Note 12: Poisson's ratio in the laboratory coordinate system

In Fig. 4c of the main text, the coordinate system is depicted, with the cantilever aligned along the y -direction in the laboratory reference frame. Stress is applied along the cantilever, corresponding to the crystal $([1\bar{1}0])$ direction with the resulting strain in the x - and z -directions scaled relative to the strain in the y -direction by Poisson's ratio.

To compute Poisson's ratio between the y and x directions ($[1\bar{1}0]$ and $[110]$) and between the y and z directions ($[1\bar{1}0]$ and $[001]$), we rotate the silicon elastic tensor to align it with the laboratory coordinate system. Specifically, we rotate the strain and stress directions by $\theta = 45^\circ$ around the z -axis. The transformation is achieved by converting the stiffness matrix from Voigt notation (C_V) to the corresponding fourth-order tensor (C_T) with the Voigt-Tensor map. The rotation is then applied using the following expression in Einstein's notation

$$C'_{T,ijkl} = R_{im}R_{jn}R_{ko}R_{lp}C_{T,mnop},$$

where R is the 3×3 rotation matrix corresponding to the rotation around the z -axis, defined as

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The rotated tensor C'_T is then transformed back to Voigt notation (C'_V) using the inverse Tensor-Voigt map. Starting from the elasticity tensor in the crystal reference frame (in GPa)

$$\begin{bmatrix} 165.7 & 63.9 & 63.9 & 0 & 0 & 0 \\ 63.9 & 165.7 & 63.9 & 0 & 0 & 0 \\ 63.9 & 63.9 & 165.7 & 0 & 0 & 0 \\ 0 & 0 & 0 & 79.6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 79.6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 79.6 \end{bmatrix},$$

we obtain the rotated stiffness matrix in the laboratory coordinate system (in GPa), as found in [15]

$$\begin{bmatrix} 194.4 & 35.2 & 63.9 & 0 & 0 & 0 \\ 35.2 & 194.4 & 63.9 & 0 & 0 & 0 \\ 63.9 & 63.9 & 165.7 & 0 & 0 & 0 \\ 0 & 0 & 0 & 79.6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 79.6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 50.9 \end{bmatrix}.$$

From this, the Poisson's ratios are computed as

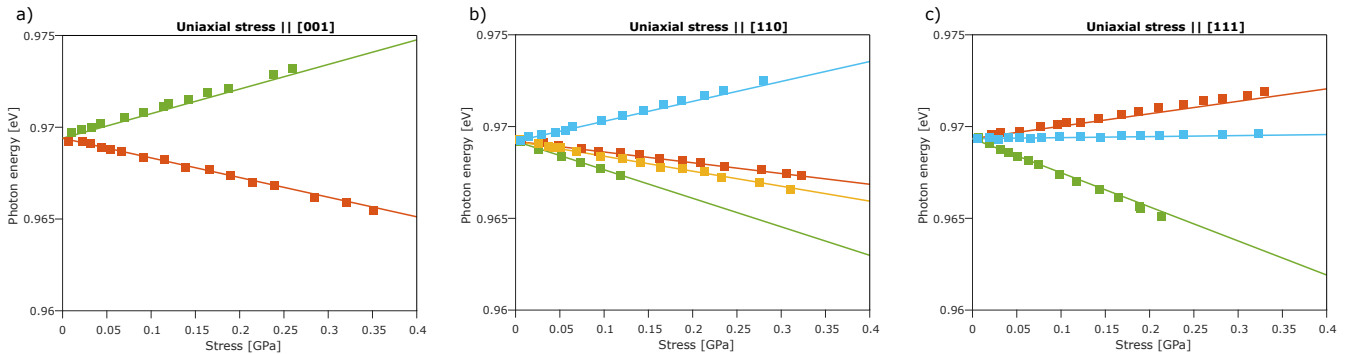
$$\nu_{y'x'} = -\frac{C_{12}}{C_{11}} = 0.0622 \quad \nu_{y'z'} = -\frac{C_{13}}{C_{11}} = 0.3617.$$

With Poisson's ratios, we obtain the perpendicular strain components to compute the effect of those components on the emitter's spectral shift using the piezospectroscopic model. The perpendicular strain components (in x and z) correspond to those found in the numerical simulations.

Supplementary Note 13: Piezospectroscopic model

A. Review piezospectroscopic model

Given the symmetry of the G-center, the point group T_d can be used to compute the rotational matrices required to transform the external strain tensor for each orientation. Starting from the identity orientation (E) referred to as defect plane (110), the symmetry operations consist of three C_2 and eight C_3 rotations, which define the corresponding rotational axes and angles. The Rodrigues' rotation formula is employed to construct these rotational matrices. By applying the matrices to the external strain tensor, initially defined in crystal coordinates, the strain response for each rotated orientation can be determined. This approach ensures that all possible strain responses for different defect orientations are accounted for. To validate this method and the piezospectroscopic model, we first reproduced the plots and fitting presented by Foy et al. [16], as shown in Supplementary Fig. 23. As part of the standard procedure, uniaxial strain was applied along three principal axes directions: [001], [110], and [111], to an ensemble of G-centers in a Si cubic sample. The fitting of the data provided the extracted value of A_p coefficients, presented in Foy's work, equal to ($A_1 = 13.4$, $A_2 = -10.7$, $A_3 = 4.8$, $A_4 = \pm 9.6$) $\frac{\text{meV}}{\text{GPa}}$. Furthermore, the number of energy splittings observed for each direction provides additional confirmation of the monoclinic-I symmetry of the defect. The reproduced plots and energy splittings align with the original results, demonstrating the consistency of the methodology. The T_d symmetry rotations were applied to determine all possible orientations and corresponding strain transformations.



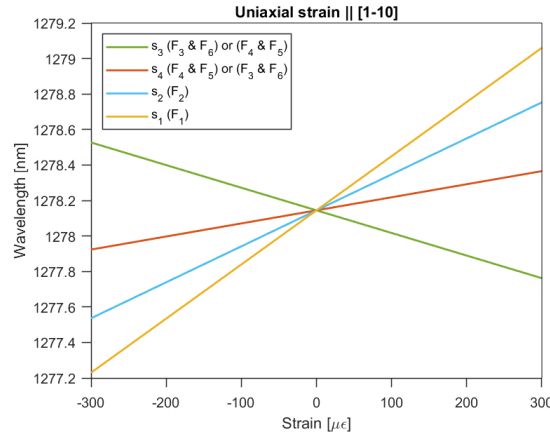
Supplementary Figure 23. Experimental data from Foy et al. [16] of uniaxial stress on G-center ensemble, piezospectroscopic model for fitting. **a)** Uniaxial stress on [001] direction, splitting into two sub-groups; **b)** Uniaxial stress on [110] direction, splitting into four sub-groups; **c)** Uniaxial stress on [111] direction, splitting into three sub-groups. The splitting represents the different stress responses for all the orientations. The initial stress is defined in crystal coordinates and then rotated for each orientation through the definition of rotational matrixes based on the T_d symmetry.

B. Piezospectroscopic shift with $[\bar{1}10]$ uniaxial strain

Once the model is validated and the rotational matrices are confirmed to be effective, the actual strain tensor must be determined. The waveguide under study is aligned along the $[\bar{1}10]$ axis, where the application of a voltage induces a deflection, generating uniaxial strain of magnitude D in the same direction. The corresponding strain components, including the vertical direction, can be calculated using FEM simulations. Assuming a crystal coordinate system where x aligns with the $[100]$ direction, y with $[010]$, and z with $[001]$, the strain tensor in crystal coordinates can be expressed as a function of D . In this configuration, the nonzero strain components are ϵ_{xx} , ϵ_{yy} , and ϵ_{xy} , with an additional ϵ_{zz} component due to the Poisson's ratio of silicon. The A_p coefficients are referred to the application of an external stress. To convert the computed strain to a stress tensor, the elastic matrix of silicon can be applied

$$\boldsymbol{\epsilon}^{\text{ext}}(D) = \begin{pmatrix} \epsilon_{xx}^{\text{ext}} \\ \epsilon_{yy}^{\text{ext}} \\ \epsilon_{zz}^{\text{ext}} \\ \epsilon_{xy}^{\text{ext}} \\ \epsilon_{yz}^{\text{ext}} \\ \epsilon_{zx}^{\text{ext}} \\ \epsilon_{xy}^{\text{ext}} \end{pmatrix} = D \cdot \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -\nu_{x,y} \\ 0 \\ 0 \\ -\frac{1}{2} \end{pmatrix}.$$

By applying the piezospectroscopic model to the computed strain tensor, we observe a splitting of the emission wavelength into four distinct responses, each characterized by a unique coefficient (slope). These responses are associated with four different shift rates under strain (s_1 , s_2 , s_3 , and s_4) extracted from the analysis shown in Supplementary Fig. 24. Specifically, the shift rates are determined as follows: $s_1 = 3.05 \times 10^{-3}$, $s_2 = 2.03 \times 10^{-3}$, $s_3 = -1.27 \times 10^{-3}$, and $s_4 = 0.74 \times 10^{-3} \frac{\text{nm}}{\mu\epsilon}$.



Supplementary Figure 24. Strain response and energy splitting derived from the piezospectroscopic model, using the FEM-extracted strain tensor along the $[\bar{1}10]$ direction. The plot shows four distinct splittings with their corresponding slopes (s_i) and rotational assignments. D is the applied strain that determines the strain components of the tensor.

This analysis further identifies six distinct classes of orientations, summarized in Supplementary Fig. 25, which group rotations resulting in equivalent final defect planes. By combining the rotational symmetry with the corresponding defect plane and strain response, we can assign a specific coefficient (shift rate of the emission wavelength as a function of uniaxial strain D) to each class.

Moreover, F_3 and F_6 have a slope s_3 assuming the negative value of A_4 , or a slope s_4 assuming positive A_4 . Vice versa for classes F_4 and F_5 . This arises from the ambiguity in determining the sign of the coefficient A_4 .

During initial studies and discoveries of G-center, experimental investigations were performed on ensembles under uniaxial strain. These studies produced multiple sets of A_p coefficients [4, 16, 17], indicating potential variations in the strain response. To address these discrepancies, the strain responses and corresponding s_i coefficients were calculated for all reported A_p sets in the literature. The average values and associated errors were then determined to account for potential variations and the corresponding error range. The final extracted coefficients are $s_1 = 2.90 \pm 0.14 \times 10^{-3}$, $s_2 = 1.96 \pm 0.07 \times 10^{-3}$, $s_3 = -1.20 \pm 0.17 \times 10^{-3}$ and $s_4 = 0.67 \pm 0.19 \times 10^{-3} \frac{\text{nm}}{\mu\epsilon}$.

Class F₁

Plane defect: (110)		
Coefficient	Average value (nm/ $\mu\epsilon$)	Relative error
s_1	2.90×10^{-3}	4.7%
Rotation	Vector	Angle
E	[0 0 0]	0°
C ₂	[0 0 1]	180°

Class F₂

Plane defect: (1-10)		
Coefficient	Average value (nm/ $\mu\epsilon$)	Relative error
s_2	1.96×10^{-3}	3.4%
Rotation	Vector	Angle
C ₂	[1 0 0]	180°
C ₂	[0 1 0]	180°

Class F₃

Plane defect: (10-1)		
Coefficient	Average value (nm/ $\mu\epsilon$)	Relative error
s_3 (for $A_s < 0$)	-1.20×10^{-3}	14.4%
s_4 (for $A_s > 0$)	0.67×10^{-3}	28%
Rotation	Vector	Angle
C ₃	[1 1 -1]	120°
C ₃	[1 -1 1]	120°

Class F₄

Plane defect: (01-1)		
Coefficient	Average value (nm/ $\mu\epsilon$)	Relative error
s_4 (for $A_s < 0$)	0.67×10^{-3}	28%
s_3 (for $A_s > 0$)	-1.20×10^{-3}	14.4%
Rotation	Vector	Angle
C ₃	[-1 -1 1]	120°
C ₃	[1 -1 -1]	120°

Class F₅

Plane defect: (101)		
Coefficient	Average value (nm/ $\mu\epsilon$)	Relative error
s_4 (for $A_s < 0$)	0.67×10^{-3}	28%
s_3 (for $A_s > 0$)	-1.20×10^{-3}	14.4%
Rotation	Vector	Angle
C ₃	[-1 1 1]	120°
C ₃	[-1 -1 -1]	120°

Class F₆

Plane defect: (011)		
Coefficient	Average value (nm/ $\mu\epsilon$)	Relative error
s_3 (for $A_s < 0$)	-1.20×10^{-3}	14.4%
s_4 (for $A_s > 0$)	0.67×10^{-3}	28%
Rotation	Vector	Angle
C ₃	[1 1 1]	120°
C ₃	[-1 1 -1]	120°

Supplementary Figure 25. Summary of all possible symmetry operations, their corresponding orientations, and the resulting defect planes. Each sub-class, defined by its final defect plane, is associated with the respective shift rate variation (slope). The identity operation (E), representing the starting configuration, corresponds to the defect plane (110). The values of the s_i coefficients are presented, with the relative error reflecting variations in the A_p coefficients reported in early studies on the G-center. For each set of A_p coefficients, the s_i values were calculated, and the average values with their associated errors are reported.

C. T-center model and comparison

This work is largely applicable, as it provides a framework for tuning other color centers in silicon. The T-center, another color center in silicon with the same point group of G-center (monoclinic C_{1h}), emits in the O-band, around 1326 nm. Unlike the G-center, the T-center is characterized by two excited states, TX_0 and TX_1 , where the strong asymmetry of the defect splits the TX state into two levels separated by 1.76 meV.

A preliminary strain model for the T-center includes the same piezospectroscopic shift common to both excited states, with an additional term that models the defect potential as internal strain, responsible for this splitting [18]. By applying the strain tensor computed in this work to the T-center model, we can perform a theoretical comparison of the energy shifts. Due to the shared symmetry, the number of splittings remains the same as for the G-center, which is equal to four. However, the strain response is nonlinear to D due to the additional term in the Hamiltonian. For the TX_0 state, the maximum shift in wavelength, $\Delta\lambda$ is ≈ 1 nm, for the F_2 class. In contrast, for the F_1 class, characterized by the defect plane (110), the TX_1 state exhibits a significantly larger strain response, shifting by almost 2 nm, while the TX_0 state shifts by ≈ 0.4 nm under a maximum applied strain of $|D| = 300\mu\epsilon$. As for the G-center, the other classes show a relatively smaller strain response.

Supplementary References

- [1] Adam N. McCaughan, Alexander N. Tait, Sonia M. Buckley, Dylan M. Oh, Jeffrey T. Chiles, Jeffrey M. Shainline, and Sae Woo Nam, “PHIDL: Python-based layout and geometry creation for nanolithography,” *Journal of Vacuum Science & Technology B* **39**, 062601 (2021).
- [2] Valeria Saggio, Carlos Errando-Herranz, Samuel Gyger, Christopher Panuski, Mihika Prabhu, Lorenzo De Santis, Ian Christen, Dalia Ornelas-Huerta, Hamza Raniwala, Connor Gerlach, Marco Colangelo, and Dirk Englund, en“Cavity-enhanced single artificial atoms in silicon,” *Nature Communications* **15**, 5296 (2024).
- [3] Mihika Prabhu, Carlos Errando-Herranz, Lorenzo De Santis, Ian Christen, Changchen Chen, Connor Gerlach, and Dirk Englund, “Individually addressable and spectrally programmable artificial atoms in silicon photonics,” *Nature Communications* **14**, 2380 (2023).
- [4] K. Thonke, H. Klemisch, J. Weber, and R. Sauer, “New model of the irradiation-induced 0.97-eV (G) line in silicon: A $C_s - Si^*$ complex,” *Physical Review B* **24**, 5874–5886 (1981).

- [5] Johannes Aberl, Enrique Prado Navarrete, Merve Karaman, Diego Haya Enriquez, Christoph Wilflingseder, Andreas Salomon, Daniel Primetzhofer, Markus Andreas Schubert, Giovanni Capellini, Thomas Fromherz, Peter Deák, Péter Udvarhelyi, Song Li, Ádám Gali, and Moritz Brehm, en“‘All-Epitaxial Self-Assembly of Silicon Color Centers Confined Within Sub-Nanometer Thin Layers Using Ultra-Low Temperature Epitaxy,” *Advanced Materials* **36**, 2408424 (2024).
- [6] Alrik Durand, Yoann Baron, Félix Cache, Tobias Herzig, Mario Khoury, Sébastien Pezzagna, Jan Meijer, Jean-Michel Hartmann, Shay Reboh, Marco Abbarchi, Isabelle Robert-Philip, Jean-Michel Gérard, Vincent Jacques, Guillaume Cassaboïs, and Anaïs Dréau, “Genuine and faux single G centers in carbon-implanted silicon,” (2024).
- [7] Peichuan Yin, John R. Serafini, Zhan Su, Ren-Jye Shiue, Erman Timurdogan, Michael L. Fanto, and Stefan Preble, “Low connector-to-connector loss through silicon photonic chips using ultra-low loss splicing of SMF-28 to high numerical aperture fibers,” *Optics Express* **27**, 24188–24193 (2019).
- [8] Jinyi Du, George F. R. Chen, Hongwei Gao, James A. Grieve, Dawn T. H. Tan, and Alexander Ling, “Demonstration of a low loss, highly stable and re-useable edge coupler for high heralding efficiency and low $g^{(2)}(0)$ SOI correlated photon pair sources,” *Optics Express* **32**, 11406–11418 (2024).
- [9] Nir Rotenberg, Pierre Türschmann, Harald R. Haakh, Diego Martin-Cano, Stephan Götzinger, and Vahid Sandoghdar, “Small slot waveguide rings for on-chip quantum optical circuits,” *Optics Express* **25**, 5397–5414 (2017).
- [10] Allen Taflove and Susan C. Hagness, *Computational Electrodynamics: The Finite-Difference Time-Domain Method, Third Edition*.
- [11] Ardavan F. Oskooi, David Roundy, Mihai Ibanescu, Peter Bermel, J. D. Joannopoulos, and Steven G. Johnson, “Meep: A flexible free-software package for electromagnetic simulations by the FDTD method,” *Computer Physics Communications* **181**, 687–702 (2010).
- [12] C. Beaufils, W. Redjem, E. Rousseau, V. Jacques, A. Yu. Kuznetsov, C. Raynaud, C. Voisin, A. Benali, T. Herzig, S. Pezzagna, J. Meijer, M. Abbarchi, and G. Cassaboïs, “Optical properties of an ensemble of G-centers in silicon,” *Physical Review B* **97**, 035303 (2018).
- [13] “Structures,” in *Microsystem Design*, edited by Stephen D. Senturia (Springer US, Boston, MA, 2001) pp. 201–238.
- [14] Gary O’Brien, David J. Monk, and Liwei Lin, “MEMS cantilever beam electrostatic pull-in model,” in *Design, Characterization, and Packaging for MEMS and Microelectronics II*, Vol. 4593 (SPIE, 2001) pp. 31–41.
- [15] Matthew A. Hopcroft, William D. Nix, and Thomas W. Kenny, “What is the Young’s Modulus of Silicon?” *Journal of Microelectromechanical Systems* **19**, 229–238 (2010).
- [16] C. P. Foy, M. C. do Carmo, G. Davies, and E. C. Lightowlers, “Uniaxial stress measurements on the 0.97 eV line in irradiated silicon,” *Journal of Physics C: Solid State Physics* **14**, L7 (1981).
- [17] G. Davies, E. C. Lightowlers, and M. do Carmo, “Carbon-related vibronic bands in electron-irradiated silicon,” *Journal of Physics C: Solid State Physics* **16**, 5503 (1983).
- [18] Chloe Clear, Sara Hosseini, Amirhossein AlizadehKhaledi, Nicholas Brunelle, Austin Woolverton, Joshua Kananayagam, Moein Kazemi, Camille Chartrand, Mehdi Keshavarz, Yihuang Xiong, Oney O. Soykal, Geoffroy Hautier, Valentin Karassiouk, Mike Thewalt, Daniel Higginbottom, and Stephanie Simmons, “Optical transition parameters of the silicon T centre,” (2024)